

Plant Disease Prediction using CNN Image Classifier

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1 Introduction

Plant diseases are one of the major challenges in agriculture and can significantly reduce crop productivity and quality. Traditional disease detection methods depend on manual inspection by agricultural experts, which is time-consuming and often unavailable in rural areas.

This project proposes a deep learning-based solution for automatic plant disease prediction using image classification techniques. The system uses Convolutional Neural Networks (CNNs) and hybrid deep learning architectures to identify healthy and diseased plant leaves from images.

2 Objective

The main objectives of this project are:

- Automatic detection of plant diseases
- Reduction of manual diagnosis effort
- Early disease identification
- Support for precision agriculture

3 Dataset Overview

The project uses the **PlantVillage Dataset** collected from Kaggle.

Total Images	54,305
Number of Classes	38
Dataset Size	2.33 GB
Dataset Source	Kaggle

Table 1: Dataset Information

Dataset Link:

<https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset>

4 Methodology

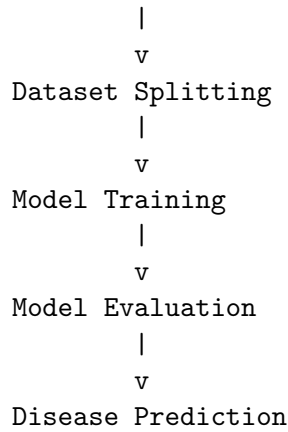
The workflow of the project is summarized below:

Dataset Collection

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Data Preprocessing



5 Data Preprocessing

The preprocessing techniques applied include:

- Image resizing to 224×224
- Pixel normalization
- Data augmentation
 - Rotation
 - Shifting
 - Zooming
 - Horizontal flipping

The dataset was divided into:

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

6 Tools and Technologies

Programming Language

- Python

Frameworks and Libraries

- PyTorch
- Torchvision
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

Development Environment

- Jupyter Notebook

7 Models Used

The following deep learning models were implemented:

- Simple CNN
- ResNet18
- ResNet50
- Vision Transformer (ViT)
- AllCNN
- ResNet18 with Attention
- CNN with Attention

8 Training Configuration

Optimizer	Adam
Loss Function	Cross Entropy Loss
Learning Rate	0.003
Epochs	5
Batch Size	32

Table 2: Training Parameters

9 Results

The models successfully classified plant diseases across multiple categories. Among all implemented models, **ResNet18** achieved the best overall performance on the test dataset.

Key outcomes:

- Accurate disease classification
- Improved feature extraction using attention mechanisms
- Strong performance across 38 disease classes
- Practical application potential in precision agriculture

10 Future Work

Future improvements may include:

- Web and mobile application deployment
- Real-time camera-based disease detection
- Lightweight model optimization
- Advanced transformer-based architectures

11 Conclusion

This project demonstrates the effectiveness of deep learning techniques for automated plant disease prediction. CNN-based and hybrid architectures can accurately classify plant diseases from leaf images and provide a valuable solution for modern agricultural systems.

— End of Report —